

```
from itertools import permutations
```

```
def tsp(graph):
```

```
    n = len(graph)
```

```
    cities = list(range(1, n))
```

```
    minimum = float('inf')
```

```
    best_path = None
```

```
    for path in permutations(cities):
```

```
        cost = 0
```

```
        current = 0
```

```
        for city in path:
```

```
            cost += graph[current][city]
```

```
            current = city
```

```
        cost += graph[current][0]
```

```
        if cost < minimum:
```

```
            minimum = cost
```

```
            best_path = (0,) + path + (0,)
```

```
    return minimum, best_path
```

```
graph = [
```

```
    [0, 10, 15, 20],
```

```
    [10, 0, 35, 25],
```

```
    [15, 35, 0, 30],
```

```
    [20, 25, 30, 0]
```

```
]
```

```
cost, path = tsp(graph)
```

```
print("Minimum Cost =", cost)
```

```
print("Best Path =", path)
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit
on win32

Enter "help" below or click "Help" above for more information.

```
>>>
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/Big-O,  $\Omega$  and  $\Theta$ .py ==
```

```
Minimum Cost = 80
```

```
Best Path = (0, 1, 3, 2, 0)
```

```
>>>
```

```
|
```